CURRICULUM VITAE
Dr Kyunghoon Han



Theoretical and computational physicist; applied mathematician; AI/ML scientist

Address 160 avenue de la Faïencerie, L-1511 Luxembourg, Luxembourg
Telephone (+352) 661 171 771
E-mail kyunghoon.h@gmail.com *or* kyunghoon.han@uni.lu
Webpage https://kyunghoon-han.github.io
GitHub https://github.com/kyunghoon-han

Patent and publication information is provided on the final page.

Name	Kyunghoon Han	**Citizenship**	Canadian
Date of birth	7 August 1990	**Gender**	Male

Education

Award date	University and diploma details
25 September 2025	**University of Luxembourg**, Luxembourg Doctor of Physics (*Docteur en Physique*)
Doctoral thesis *Analysis, Calculation, and Understanding of Molecular Vibrations: Decomposition of Empirical Spectra, Simulation Targeted to a Frequency Band, and the Theoretical Framework of Oscillatory Motion as Fat Manifolds.* Supervisor: Dr Joshua T. Berryman.	
6 December 2017	**Université de Tours**, France Master of Science, Technologies and Health (*Master de Sciences, Technologies, Santé*) Physical Sciences and Imaging (*Mention Sciences Physiques et Imagerie*) Specialty : Nonlinear models in Physics (*Spécialité Modèles non-linéaires en physique)*
Master's thesis *Simulation of Vortex Behaviour in the Inner Crust of a Neutron Star.* Supervisor: Dr Yves Lansac.	
13 June 2014	**University of Waterloo**, Canada Bachelor of Mathematics Honours Pure Mathematics and Mathematical Physics (double major)
Undergraduate thesis *Osmotic Compaction of Bacterial Chromosome* (PHYS 437A/B). Supervisor: Prof. Bae-Yeun Ha.	
June 2008	**Colonel Gray Senior High School**, Charlottetown, Prince Edward Island, Canada

Fields of Specialisation

Computational physics	Molecular dynamics, molecular vibrations and spectroscopy; statistical, nonlinear, and stochastic physics; continuum, fluid, granular, and relativistic dynamics.
Scientific machine learning	Machine-learned force fields; physics-constrained neural representations and closures; surrogate modelling; quantitative evaluation of accuracy, stability, robustness, and computational efficiency.
Artificial intelligence	Multimodal generative AI spanning language, speech, facial animation, image, and video systems; creation of deployable digital personalities, including RuiCovery.
Applied mathematics	Dynamical systems, chemical reaction networks, category theory, information theory and partial information decomposition, and Euclidean/non-Euclidean geometry.

Current Research

Chemical-Reaction-Network Formulation of Continuum Mechanics. Developing a mathematical formulation of continuum mechanics based on chemical reaction networks (CRNs), in which macroscopic fluid motion emerges from structured local interactions. Implemented computationally efficient, compact neural representations that reproduce three-dimensional Navier–Stokes flow with velocity discrepancies on the order of 10^{-4} m/s in periodic water-flow benchmarks. The framework offers a physically interpretable alternative to conventional black-box flow prediction and provides a foundation for extending learned continuum dynamics to granular-flow systems.

Professional Experience

Period	Position and institution	Principal work
Jan.–July 2026	**Postdoctoral Researcher** University of Luxembourg	Developed and benchmarked machine-learned force fields and frequency-domain molecular dynamics, with reproducible large-scale simulation and analysis workflows.
Sept. 2021– Dec. 2025	**Doctoral Researcher** University of Luxembourg	Researched molecular vibrations and spectroscopy; developed the open-source TIHI Toolkit and computational pipelines using ORCA, AMBER, Python, and HPC systems.
Apr. 2019– June 2021	**Head of Research / AI Scientist** SRuniverse, Seoul	Led a team developing and productionising multimodal NLP, TTS/STT, facial-image, lip-sync, and video-generation systems; created RuiCovery, an AI-driven YouTube personality.
Oct. 2017– Apr. 2018	**Researcher** Hankook Life Science Institute, Seoul	Conducted rodent-anatomy research and contributed to public-facing physics education.
Summers 2016 and 2018	**Lecturer** Steven Academy, Seoul	Taught physics, chemistry, mathematics, Latin, and English to students from secondary-school to post-undergraduate level.

Period	Position and institution	Principal work
Apr.–Oct. 2015	**Technical Translator** Hankook Tire, Daejeon	Translated Korean technical and manufacturing documentation into English for use at Hankook Tire's Tennessee plant.
Sept. 2010– Apr. 2011	**Research Assistant** Canada Centre for Remote Sensing, Natural Resources Canada, Ottawa	Developed RADARSAT-2 signal-processing and orbit-interpolation tools, including Doppler and relativistic image corrections using Canadian digital elevation model.
Jan.–Apr. 2010	**Student Intern** Kwangwoon University RFIC Centre, Seoul	Designed and analysed a 9.1 GHz band-stop gapped-ring resonator, contributing to a KIEES conference paper.

Teaching Experience

All teaching listed below was delivered in English to international or non-native English-speaking students.

Level	Institution / role	Courses taught	Audience and scope
Master's level	**University of Luxembourg** Principal instructor, 2024	Computational Methods for Physicists	Graduate physics students; numerical methods, Python scientific computing, simulation, machine learning, quantum computing, and data analysis.
Undergraduate level	**University of Luxembourg** Teaching assistant, 2022–2023	Data Science and Machine Learning in Physics, Programming for Physics	Undergraduate physics students; programming, data analysis, and machine-learning foundations.
Secondary / university-entry	**Steven Academy**, Seoul Private-school teacher, 2016 and 2018	GRE Physics and Chemistry; AP Physics, Chemistry, Calculus, Latin, and Environmental Science; IB Physics, Mathematics, and Chemistry; SAT Mathematics; high-school English	International secondary-school, university-entry, and ESL students.

Technical Profile

Scientific ML	PyTorch and JAX; SO3LR and other machine-learned force fields; physics-constrained neural models and learned closures; dataset construction; training and inference; accuracy, stability, robustness, and performance benchmarking.
Generative AI systems	NLP; text-to-speech; speech-to-text; Griffin–Lim vocoding; facial-expression synthesis, lip-sync, image generation, and video generation; RuiCovery digital-personality pipeline.
Simulation and HPC	Molecular dynamics and vibrational spectroscopy; ORCA and AMBER; continuum, fluid, granular, and relativistic simulation; CPU/GPU execution, checkpointed long runs, and heterogeneous cluster workflows.
MLOps and deployment	Docker, Kubernetes, Jenkins; modular training/inference services; experiment tracking, model registries, automated validation, observability, and controlled model comparison.
Software engineering	Python, C/C++, Julia, Haskell, JavaScript, WebAssembly, LaTeX, Git; reusable packages, command-line tools, notebooks, reproducible releases, and automated scientific data, figure, and video pipelines.

Awards and Leadership

September 2008	University of Waterloo President's Scholarship.
2007	Gold Medal, Fermat Mathematics Contest.
Sept. 2012– Apr. 2013	President, University of Waterloo Space Society.

Languages and International Experience

Languages	English and Korean (fluent); French (professional working proficiency); Japanese (A2); German (B1.1); Luxembourgish (A1, currently studying); Ancient Egyptian (Hobby, A1-level).
International experience	Education, research, teaching, and industry experience in Canada, Republic of Korea, France, and Luxembourg.

LIST OF RESEARCH ACHIEVEMENTS

Kyunghoon Han

A. Patent

- K. Han, *Method and Apparatus for Automated Evaluation of Synthetic Speech Based on Artificial Intelligence*. Republic of Korea Patent Application No. 10-2020-0162905; Publication No. 10-2022-0074415. Applicant: SRuniverse. DOI: 10.8080/1020200162905.

B. Publications and Scholarly Works

- **2026.** S. Suárez-Dou, M. Gallegos, **K. Han**, F. N. Brünig, J. T. Berryman, and A. Tkatchenko*, "Quantum-Accurate Vibrational and Conformational Dynamics from Small Molecules to Proteins with Machine-Learned Force Fields." Accepted for publication in *Nature Communications*, 15 October 2026, manuscript #NCOMMS-26-004349C; arXiv:2601.09845.
- **2026. K. Han***, A. Tkatchenko, and J. T. Berryman, "Symplectic and Thermodynamically Consistent Molecular Dynamics in the Frequency Domain." Under revision at *Physical Review Letters* following peer review; revised manuscript and response to referees submitted. arXiv:2606.11809.
- **2026. K. Han***, "Category Theoretic Framework for Chemistry I: A Tower of Chemistry." Submitted to the *Journal of Mathematical Chemistry*. arXiv:2606.25774.
- **2026. K. Han***, "Categorification of Chemical Reactions: A Bottom-Up Tower from Stoichiometry to Quantum Structure." arXiv:2605.14974.
- **2026. K. Han*** and M. Gallegos, "Partial Information Decomposition of Electronic Observables Along a Reaction Coordinate." arXiv:2603.05672.
- **2025.** S. Dey, E. H. Field, Y. Wang, **K. Han**, R. Kumar, L. J. W. Shimon, S. Liu, S. Guerin, J. T. Berryman, W. Ji*, and S. Bera*, "Fluorination Induced Inversion of Helicity and Self-Assembly Into Cross-α Like Piezoelectric Amyloids by Minimalistic Designer Peptide." *Small* **21**(18), 2500288 (2025).
- **2025. K. Han**, *Analysis, Calculation and Understanding of Molecular Vibrations: Decomposition of Empirical Spectra, Simulation Targeted to a Frequency Band, and Theoretical Framework of Oscillatory Motion as Fat Manifolds*. Doctoral thesis, University of Luxembourg. Handle: 10993/66341.
- **2024. K. Han***, A. Boziki, A. Tkatchenko, and J. T. Berryman, "TIHI Toolkit: A Peak Finder and Analyzer for Spectroscopic Data." *ACS Omega* **9**(50), 49397–49410 (2024).
- **2022.** M. Charnley, S. Islam, G. K. Bindra, J. Engwirda, J. Ratcliffe, J. Zhou, R. Mezzenga, M. D. Hulett, **K. Han**, J. T. Berryman*, and N. P. Reynolds*, "Neurotoxic Amyloidogenic Peptides in the Proteome of SARS-CoV-2: Potential Implications for Neurological Symptoms in COVID-19." *Nature Communications* **13**(1), 3387 (2022).
- **2022.** A. Fallani, L. Medrano Sandonas, **K. Han**, and A. Tkatchenko*, "Towards the Inverse Design of Molecules with Targeted Quantum-Mechanical Properties." *APS March Meeting Abstracts*, Vol. 2022, T49-013 (March 2022).
- **2015. K. Han***, "On a Stochastic Construction of the Kinematics in Discrete Space-Time." *Canadian Journal of Physics* **93**(5), 496–502 (2015).
- **2010. K. Han**, N.-Y. Kim, and B. Shrestha*, "Design of 9.1 GHz Band Stop Type Gapped Ring Resonator." *KIEES Spring Conference* (June 2010).

C. Invited Talks and Conference Presentations

- K. Han, "Categorification of Chemical Reactions: From Stoichiometry to Quantum Structures." Chemical Reaction Networks in Hawai'i 2026, Honolulu, USA (May 2026).
- K. Han, "Partial Information Decomposition of Electronic Observables Along Reaction Pathways." Korean Scientists and Engineers Association in Luxembourg, 1st Open Seminar, Luxembourg (April 2026, **Invited**).
- K. Han, "On Information-Theoretic Re-definition of Chemical Bonds Using Partial Information Decomposition." German Physical Society (DPG) Spring Meeting, Dresden, Germany (March 2026).
- K. Han, "On Quantum Radars and Partial Information Decomposition." IEEE Aerospace and Electronic Systems Society, Ottawa Chapter, Ottawa, Canada (November 2025, **Invited**).

*Asterisks reproduce the corresponding-author markings in the source achievement record.